

Undifferentiated abdominal pain / shock

Section 1: Case Summary

Scenario Title:	Undifferentiated abdominal pain and shock
Keywords:	Abdominal pain, shock, perforated viscus
Brief Description of Case:	A 67yr old male with multiple comorbidities is brought by ambulance with a 3-day history of diffuse abdominal pain. The history is vague and the differential of his symptoms remains very broad. He develops significantly worsening pain and hypotension and becomes obtunded. As the patient's condition deteriorates, the team must initiate management of abdominal pain plus shock and support the hemodynamics with vasopressors/inotropes. The team will need to intubate to facilitate advanced imaging and definitive care.

Goals and Objectives	
Educational Goal:	To investigate and manage a patient undifferentiated abdominal pain and shock
Objectives: (Medical and CRM)	- Maintain a broad differential diagnosis for abdominal pain and shock - Order appropriate investigations for undifferentiated abdominal pain including utilization of bedside ultrasound - Appropriately manage hemodynamics in the setting of shock - Recognize and manage a physiologically difficult airway - Establish clear leadership - Delegate roles appropriately and maintain a shared mental model with verbal summaries
EPAs Assessed:	N/A

Learners, Setting and Personnel	
Target Learners:	☒ Junior Learners ☒ Senior Learners ☐ Staff
	☒ Physicians ☒ Nurses ☐ RTs ☐ Inter-professional
	☐ Other Learners:
Location:	☒ Sim Lab ☐ In Situ ☐ Other:
Recommended Number of Facilitators:	Instructors: 1
	Sim Actors: 0
	Sim Techs: 1

Scenario Development	
Date of Development:	June 23, 2021
Scenario Developer(s):	Dr. Larissa Hattin
Affiliations/Institutions(s):	UBC Emergency Medicine Residency Program
Contact E-mail:	Larissa.hattin@medportal.ca
Last Revision Date:	
Revised By:	
Version Number:	1



Undifferentiated abdominal pain / shock

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Daniel Evans		Age: 73		Gender: male	Weight: 90kg
Presenting complaint: Abdominal pain x 3 days					
Temp: 37.9	HR: 118	BP: 112/76	RR: 28	O ₂ Sat: 96%	FiO ₂ : R/A
Cap glucose: 8.6 mmol/L			GCS: 15 (E4 V5 M6)		
Triage note: 67M with multiple comorbidities, here today with 3 days of gradually worsening abdo pain. Vomited x 2 at home. Hx of ETOH. Brought by EMS.					
Allergies: Sulfa drugs					
Past Medical History: - HTN - Dyslipidemia - Smoker (5 cigarettes/day) - Drinks 4 beers/day x years - Umbilical hernia - Obese			Current Medications: - Ramipril - Atorvastatin - ASA		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	
<u>Patient</u> **History vague, just wants you to "fix him" and reluctant to provide additional information - The pain started 3 days ago. It felt like a dull ache, until this morning when it became a lot worse. - The pain is mainly epigastric but now has spread to his whole abdomen. - Vomited x 2 this am. If they ask what was in it, he responds by saying "What, you think I looked at it? I was in too much pain!" - Can't provide information on stool habits - Last drink 2 days ago because his pain was too strong - Has been taking regular acetaminophen and ibuprofen, the max recommendations on the bottle for 3 days	
B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal, no edema to extremities	Neuro: Normal mental status. Moaning in pain. No tremor. No signs of withdrawal.
Resp: Shallow breaths due to abdo pain. No crackles/wheeze. Slightly decrease AE at bilateral bases.	Head & Neck: Nil acute



Undifferentiated abdominal pain / shock

Abdo: Umbilical hernia soft, easily reduced. No stigmata of liver disease. Abdo is diffusely tender with guarding. No masses. **only provide DRE if they state they would perform one: Positive FOBT.	MSK/skin: No rashes, no jaundice/scleral icterus
Other:	

Undifferentiated abdominal pain / shock

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>): adult
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Crash cart with medications Airway cart with RSI drugs Glidescope/CMAC
C. Required Medications
Analgesia: Fentanyl Ionotropes/Vasopressors: norepinephrine/epinephrine/vasopressin infusions, phenylephrine IV push doses RSI medications: ketamine IV, rocuronium IV Blood for transfusion Broad spectrum antibiotics such as Pip-Tazo
D. Moulage
Patient gown
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A: Patent, patient speaking. Moans intermittently in pain B: Shallow breathing due to abdominal pain. Decreased AE at bases C: Tachycardic, normal pulses. Abdomen diffusely tender with guarding. Umbilical hernia soft, easily reduced D: GCS 15 E: nil

Undifferentiated abdominal pain / shock

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)

Undifferentiated abdominal pain / shock

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 118 BP: 112/76 RR: 28 O ₂ SAT: 94% T: 37.9 °C GCS: 15	Patient is alert, appears in pain. Moans, vague history.	<u>Expected Learner Actions</u> ☐ Cardioresp monitors ☐ Supplemental O ₂ ☐ Interview patient ☐ Perform physical exam ☐ IV access x 2 ☐ Order labs, ECG, portable CXR/AXR ☐ Give 1L RL bolus ☐ Provides analgesia		<u>Modifiers</u> - patient settles slightly if given analgesia - O ₂ sats change to 96% with supplemental O ₂ by NP - patient becomes more aggressive if no analgesia is given <u>Triggers</u> - 5 minutes → 2. Return of investigations
2. Return of Investigations Rhythm: Sinus HR: 124 BP: 103/58 RR: 34 O ₂ SAT: 94% T: 38.1 °C GCS: 13 (E3 V4 M6) Glucose: normal	Patient is in more pain, moaning consistently. Very shallow, fast breathing. Repeat physical exam: abdomen is distended and peritonitic. eFAST: negative for free fluid. Lung sliding bilaterally. No pericardial effusion. Grossly normal contractility. IVC collapses with inspiration. Lab results are now available.	<u>Expected Learner Actions</u> ☐ Re-check vitals, including temperature and glucose ☐ Repeat physical exam ☐ Perform e-FAST exam ☐ Order VBG and blood cultures ☐ Start empiric antibiotics ☐ Gives 2 nd 1L bolus ☐ Verbalize differential and treatment plan to the team ☐ Order 2-4U of unmatched PRBC's		<u>Modifiers</u> - Lab calls with critically low Hb results and suggests ordering PRBCs - Patient has worsening hypotension if no fluids or antibiotics are initiated - BP improves slightly with 2 nd fluid bolus, improves if given blood <u>Triggers</u> - 5 minutes, or all actions complete → 3. Decline



Undifferentiated abdominal pain / shock

3. Decline Rhythm: Sinus HR: 125-135 BP: 92/45 RR: 34 O ₂ SAT: 94% T: 38.1 °C GCS: 6 (E1 V1 M4)	Patient becomes obtunded and unresponsive. Weak peripheral pulses	<u>Expected Learner Actions</u> ☐ Re-check vitals ☐ Start vasopressor ☐ Give 2U of blood ☐ Prepare for intubation: discuss plan and medications including plan for peri-intubation hypotension	<u>Modifiers</u> - BP worsens if vasopressor not started - BP improves to SBP ~105 with inotropes/blood/fluids <u>Triggers</u> - all actions complete and verbalizes intubation plan → 4. Intubation	
4. Intubation HR: 125-135 BP: 105/55 RR: 26 O ₂ SAT: 97% T: 38.1 °C GCS: 6 (E1 V1 M4) Glucose: normal		<u>Expected Learner Actions</u> ☐ Intubate patient ☐ Repeat vitals ☐ Recognize and treat post intubation hypotension if present ☐ Post intubation sedation ☐ Order STAT CT abdomen/pelvis ☐ Consult GenSx and ICU	<u>Modifiers</u> - patient will have significant post-intubation hypotension (85/40) if hemodynamics not optimized prior to intubation - patient will NOT arrest <u>Triggers</u> - patient stable enough for CT, ICU/Gen Sx consulted ends the case	

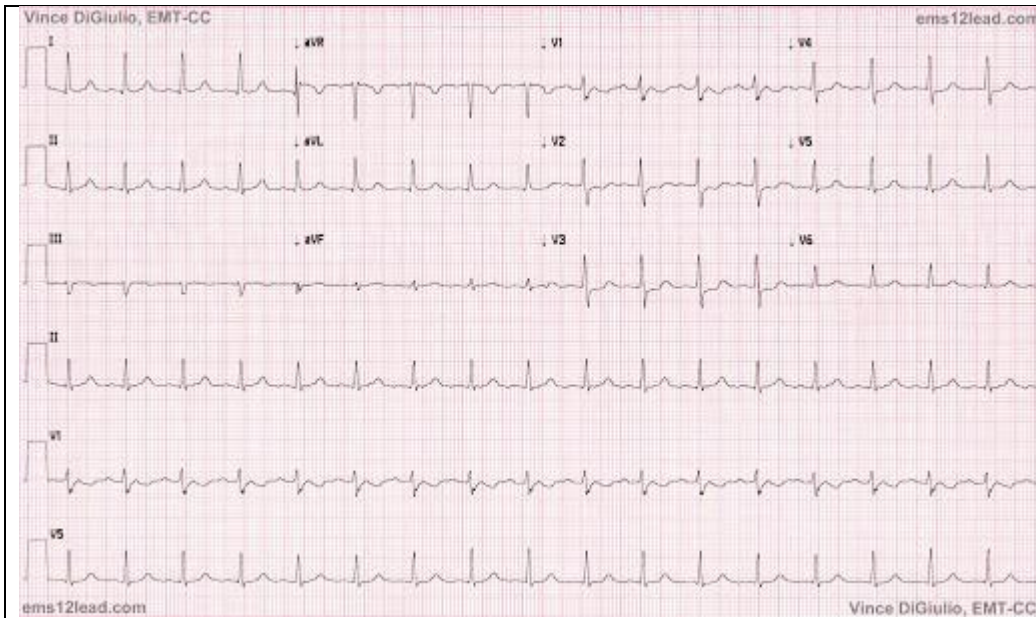
Undifferentiated abdominal pain / shock

Appendix A: Laboratory Results (all SI units)

<u>CBC</u> WBC: 17.3 Hgb: 64 Plt: 278 <u>Lytes</u> Na: 132 K: 3.2 Cl: 101 HCO ₃ : 12 AG: 19 Urea: 38 Cr: 176 GFR: 32 Glucose: 6.9 <u>VBG</u> pH: 7.27 pCO ₂ : 27 pO ₂ : 93 HCO ₃ : 12 Lactate: 3.9	<u>Cardiac/Coags</u> Trop: 0.10 INR: 1.2 aPTT: 30 sec <u>Biliary</u> AST: 204 ALT: 298 GGT: 86 ALP: 79 Bili: 14 Lipase: 302 <u>Tox</u> EtOH: 0 ASA: 0 Tylenol: Therapeutic level
---	--

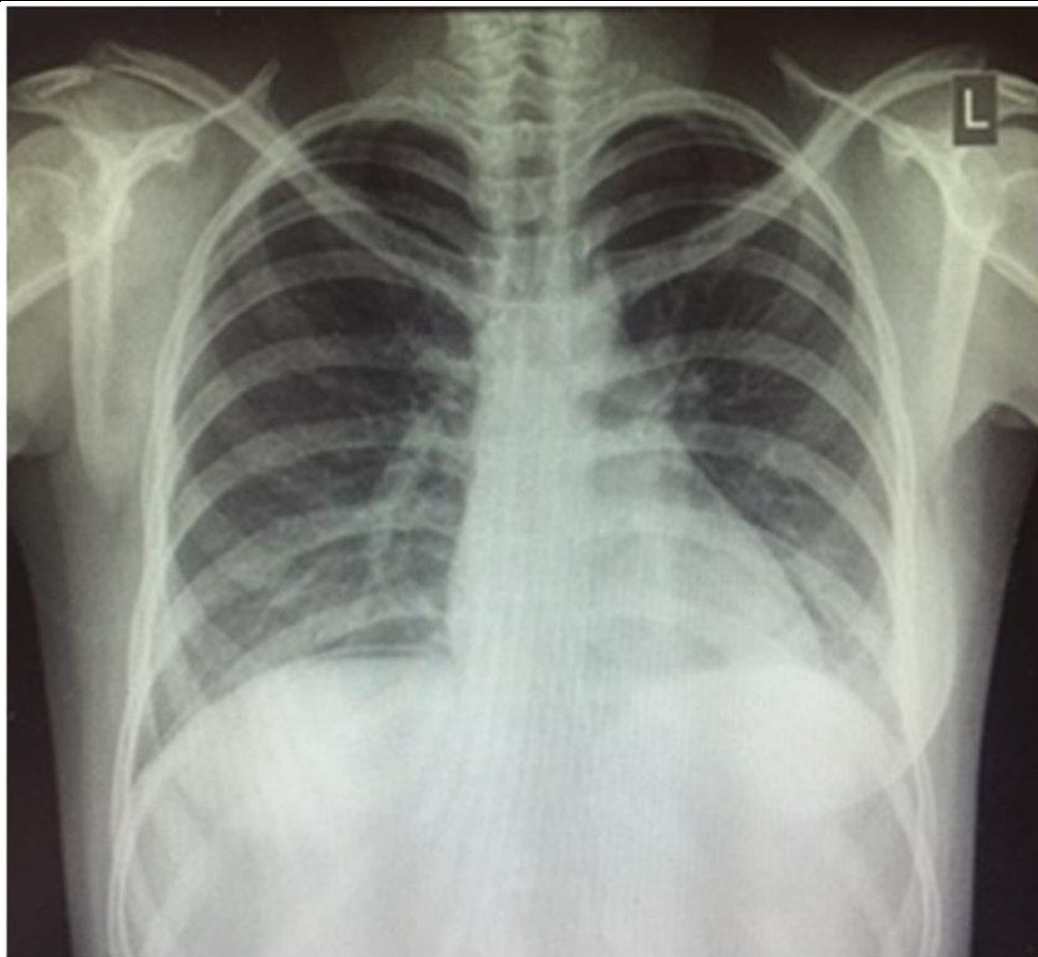
Undifferentiated abdominal pain / shock

Appendix B: ECGs, X-rays, Ultrasounds and Pictures



<http://ems12lead.com/2015/12/26/12-rhythms-of-christmas-sinus-tach/>

Undifferentiated abdominal pain / shock



https://www.researchgate.net/figure/Erect-chest-x-ray-demonstrating-subdiaphragmatic-free-air_fig1_323084315

Undifferentiated abdominal pain / shock

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Medical Debriefing Questions

- Can someone give a summary of the case?
- What is the differential diagnosis for this patient's specific presentation?
- What is meant by physiologically difficult airway?
- How would the case have changed if you prioritized getting a CT scan?
- What are the Surviving Sepsis guidelines? What is included in the 1hr bundle?

CRM Debriefing Questions

- Does everyone in the team feel like they knew what was going on in the case?
- How do you avoid anchoring bias?
- How do you manage diagnostic uncertainty in a critically ill patient?
- How did you feel when the patient was unable to provide you with information and was mad at you? How did you deal with that feeling?
- What do you feel went well during this case?
- What aspects of this case were challenging and how did you overcome them?
- How does everyone feel after that case?
- What can make it difficult to call for help, or a consultant, early? IE before you've gotten all of your investigations back?

References

1. Vakil, N. (2020). Overview of complications of peptic ulcer disease. *UpToDate*. Retrieved June 8, 2021, from https://www.uptodate.com/contents/overview-of-complications-of-peptic-ulcer-disease?search=perforated%20peptic%20ulcer§ionRank=1&usage_type=default&anchor=H365487672&source=machineLearning&selectedTitle=1~108&display_rank=1#H56805017.
2. Howell MD, Davis AM. Management of Sepsis and Septic Shock. *JAMA*. 2017;317(8):847–848. doi:10.1001/jama.2017.0131
3. De Backer D, Dorman T. Surviving Sepsis Guidelines: A Continuous Move Toward Better Care of Patients With Sepsis. *JAMA*. 2017;317(8):807–808.

